KOLLIMARLA SANJAY

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with **3+ years** of experience in designing and building scalable backend systems using **Node.js**, **TypeScript**, and **JavaScript**. Skilled in architecting multi-tenant SaaS applications, developing distributed backend services, and optimizing database performance. Proficient in designing asynchronous processing pipelines and building real-time applications, with a strong foundation in Data Structures & Algorithms, System Design, and Performance Optimization.

WORK EXPERIENCES

Software Engineer | Full-time — Feb 2024 - Present
NexusIQ Solutions — *On-site · Hyderabad, Telangana, India*

- Designed and implemented a **multi-tenant backend architecture** serving multiple organizations with secure tenant isolation and configurable authorization.
- Architected **RESTful backend services** using Node.js and TypeScript, supporting concurrent exam workflows with role-based access control across **40+ permission types**.
- Optimized MongoDB queries through indexing, aggregation pipeline improvements, and schema redesign, **reducing API latency** for high-volume operations.
- Built asynchronous processing pipelines for AI-assisted question generation and evaluation, **improving content generation efficiency by over 70%**.
- Integrated AWS S3 storage for scalable media management with secure upload/download workflows.
- Owned features through architecture, implementation, deployment, production support, and performance optimization.

Technologies / Skills Used : *Node.js, TypeScript, MongoDB, AWS S3, WebSockets, Microservices, Next.js, Framer-motion, SSE, Webhooks, Tailwind CSS, Express, Redis, Rest APIs, Docker, Queuing Systems, BullMQ*

Junior Software Engineer | Full-time — Nov 2022 - Apr 2023
Upgrad Insofe — *On-site · Hyderabad, Telangana, India*

- Built Angular components and Node.js APIs for analytics dashboards, **reducing page load time by 30%** with lazy loading.
- Collaborated with data engineering teams to integrate machine learning models into production APIs.

Technologies / Skills Used : *Angular, Node.js, Bootstrap, Express, MongoDB*

PROJECTS

Production-grade Email Queue Platform | Github Live Demo
Architect

- Architected a distributed email processing system using BullMQ and Redis, handling **10,000+ jobs/day** under load-tested workloads.
- Implemented retries, exponential backoff, DLQs, and idempotency to improve failure recovery across **4 core reliability mechanisms**.
- Developed horizontally scalable workers with priority queues, rate limiting, and SSE/webhook tracking, supporting **3 job-processing capabilities** and near-real-time status updates.

Technologies / Tools Used : *Node.js, TypeScript, BullMQ, Redis, Docker*

Distributed Rate Limiting Service
Developer

- Engineered a centralized **Redis-based rate limiter** supporting **1,000+ requests/sec** with shared rate-limit state across distributed application instances.
- Implemented Token Bucket and Sliding Window algorithms using atomic Redis operations and Lua scripts to prevent race conditions.
- Built reusable **Express middleware** supporting **100+ concurrent clients**, configurable throttling policies, and per-client rate limits.

Technologies / Tools Used : *Node.js, TypeScript, Redis, Lua, Express*

TinyURL - Distributed URL Shortening Service
Developer

- Architected a scalable URL shortening service targeting **1,000+ redirects/sec**, featuring unique short-link generation, analytics, and configurable expiration policies.
- Optimized redirect performance using Redis caching and MongoDB, achieving **sub-50 ms p95 response latency** under load testing.
- Implemented asynchronous click analytics, duplicate detection, and automated link expiration to support high-volume redirect traffic and reliable lifecycle management.

Technologies / Tools Used : *Node.js, TypeScript, MongoDB, Redis, Docker, AWS*

SKILLS

Databases :	MongoDB, MySQL, Redis
Frameworks & Libraries :	Node.js, Express.js, React.js, Angular
Programming Languages :	Java, TypeScript, JavaScript, SQL, Python, programming
Soft Skills :	Data Structures & Algorithms, System Design, Distributed Systems, Database Design, Caching, Concurrency, Event-Driven Architecture, RBAC, Design Patterns
Tools & Platforms :	AWS S3, Docker, Git, Postman, Webpack, CI/CD, CI/CD tools, Kubernetes, AWS EKS, GCP GKE, Azure AKS

EDUCATIONS

Year	Degree	Institute	CGPA
2019 - 2023	Bachelor of Technology in Computer Science & Engineering	Geethanjali College of Engineering and Technology	7.8/10

CERTIFICATIONS

- Full Stack Development (React, Node.js) - NxtWave
- Java Full Stack - Wipro TalentNext
- Machine Learning using Python - Verzeo
- Git, SQL, Command Line & Responsive Design - NxtWave